Short Syllabus

BCSE335L Healthcare Data Analytics (3-0-0-3)

Introduction to Healthcare Data Analytics - Need for Healthcare Analytics , Foundations of Healthcare Analytics; Healthcare Foundations - Healthcare delivery, Healthcare financing, Healthcare policy; Machine Learning Foundations for Healthcare - Model frameworks for medical decision making, Tree-like reasoning, Probabilistic reasoning; Measuring Healthcare Quality - Medicare value-based programs; Making Predictive Models in Healthcare - Obtaining and Importing the NHAMCS Dataset, Making the Response Variable; Healthcare Analytics Applications - Descriptive Analytics Applications, Predictive Analytics Applications; Healthcare and Emerging Technologies- Healthcare and the Internet of Things; Healthcare and Emerging Technologies - Healthcare analytics and social media.

Course code	Course Title	L	T	P	C
BCSE335L	**Healthcare Data Analytics**	**3**	**0**	**0**	**3**
Pre-requisite	**NIL**	**Syllabus version**			
		1.0			

Course Objectives

1. Describe how data-based healthcare can help in improving outcomes for patient health.
2. To design data models that combine patient records from multiple sources to form a patient centric view of data.
3. To use data analytics to find health concerns and solutions to the problem faced by a patient.
4. To find meaningful patterns and trends in healthcare data to help the overall population.

Course Outcomes

At the end of the course, the student will be able to

1. Explain the concepts of Healthcare Data Analytics and healthcare foundations.
2. Apply machine learning techniques on healthcare data analytics.
3. Measure and analyse the quality of health-care systems.
4. Develop models for effective predictions in healthcare applications.
5. Use modern day emerging technologies in healthcare data analytics process.

Module:1	**Introduction to Healthcare Data Analytics**	**3 hours**

Introduction – Need for Healthcare Analytics - Foundations of Healthcare Analytics – Examples of Healthcare Analytics.

Module:2	**Healthcare Foundations**	**5 hours**

Healthcare delivery - Healthcare financing - Healthcare policy – Handling Patient data: the journey from patient to computer - Standardized clinical codesets - Breaking down healthcare analytics: population, medical task, data format, disease.

Module:3	**Machine Learning Foundations for Healthcare**	**8 hours**

Model frameworks for medical decision making: Tree-like reasoning, Probabilistic reasoning and Bayes theorem, Criterion tables and the weighted sum approach, Pattern association and neural networks - Machine learning pipeline: Loading the data, Cleaning and preprocessing the data, Exploring and visualizing the data, Selecting features, Training the model parameters, Evaluating model performance.

Module:4	**Measuring Healthcare Quality**	**8 hours**

Introduction to healthcare measures, Medicare value-based programs: The Hospital Value-Based Purchasing (HVBP) program, The Hospital Readmission Reduction (HRR) program, The Hospital-Acquired Conditions (HAC) program, The End-Stage Renal Disease (ESRD) quality incentive program, The Skilled Nursing Facility Value-Based Program (SNFVBP), The Home Health Value-Based Program (HHVBP), The Merit-Based Incentive Payment System (MIPS).

Module:5	**Making Predictive Models in Healthcare**	**8 hours**

Introduction to Predictive Analytics – Obtaining and Importing the NHAMCS Dataset – Making the Response Variable - Splitting the Data into Train and Test Sets - Preprocessing the Predictor Variables – Building the Models – Using the Models to Make Predictions – Improving our Models.

Module:6	**Healthcare Analytics Applications**	**6 hours**

Introduction - Descriptive Analytics Applications - Predictive Analytics Applications - Prescriptive Analytics Application.

Module:7	**Healthcare and Emerging Technologies**	**5 hours**

Healthcare analytics and the internet - Healthcare and the Internet of Things - Healthcare

analytics and social media - Healthcare and deep learning - Obstacles, ethical issues, and limitations.		
Module:8	**Contemporary Issues**	**2 hours**
	Total Lecture hours	**45 hours**
Text Book(s)		
1.	Kumar, Vikas Vik. Healthcare Analytics Made Simple: Techniques in healthcare computing using machine learning and Python. Packt Publishing Ltd, 2018.	
2.	El Morr, Christo, and Hossam Ali-Hassan. Analytics in healthcare: a practical introduction. Springer, 2019.	
Reference Books		
1.	Dinov, Ivo D. "Data Science and Predictive Analytics." Springer, Ann Arbor, MI, USA https://doi. org/10 1007 (2018): 978-3.	
2.	Yang, Hui, and Eva K. Lee, eds. Healthcare analytics: from data to knowledge to healthcare improvement. John Wiley & Sons, 2016.	
Mode of Evaluation: CAT / written assignment / Quiz / FAT / Project / Seminar / group discussion		

Recommended by Board of Studies	12-05-2022		
Approved by Academic Council	No. 66	Date	16-06-2022